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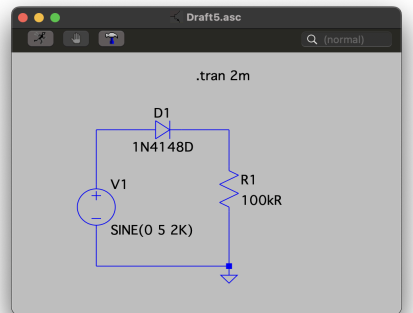
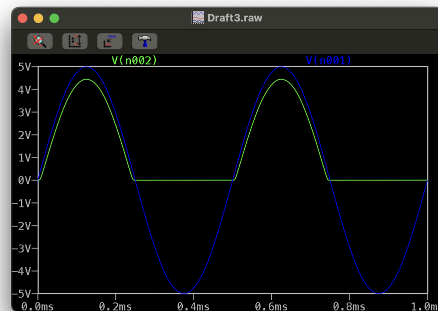
ECE 65

Pre-Lab 4- Diode Waveform Shaping Circuits

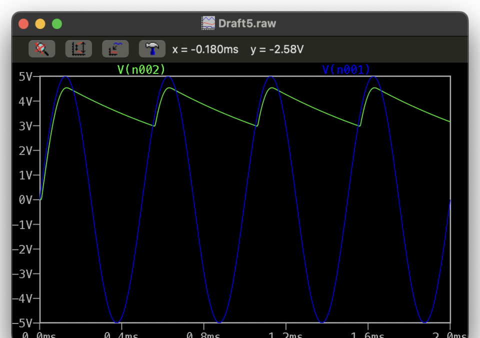
Experiment 1:

1)



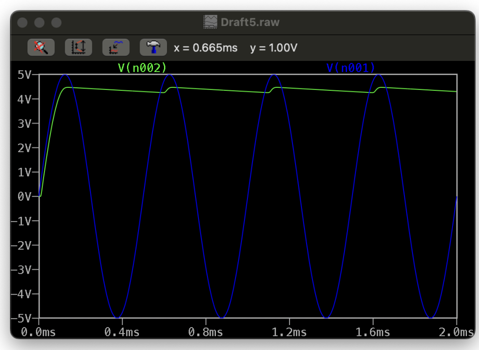


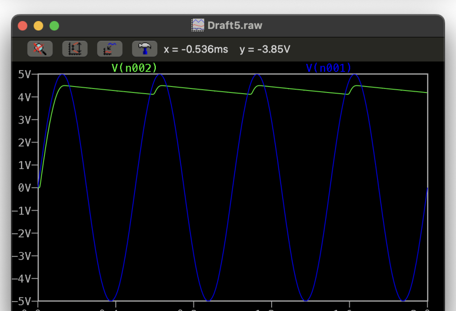
2 and 3)



← Figure 1 (10nF)

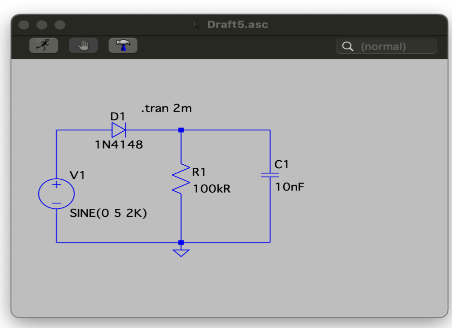
Figure 2 (90nF) →





← Figure 3 (50nF)

Figure 4
(Schematic) →

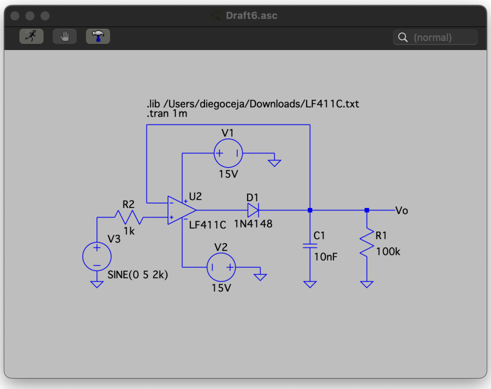


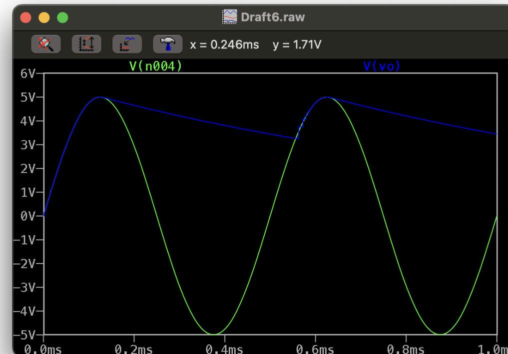
The 90nF (**Figure 2**) is the best peak detector, the mediocre is 50nF (**Figure 3**), and the worst one is 10nF (**Figure 1**). I just chose the numbers by random.

4) I would say it does since it makes sense that the 90nF is the best one since it holds the peak values of the input signal longer and has minimal discharge between peaks, making it more stable compared to like 10nF where it is unstable and holds the input signal very short.

Experiment 2:

1)

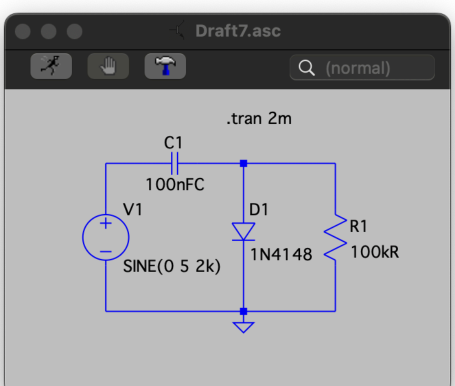




2) The output voltage from the circuit reflects the highest peak value of the input signal until it is gradually discharged through the resistor, which is why we see it going down in every peak. The circuit from experiment 1 and this circuit are very similar, they both work the same in terms of holding the input signal for a very short time.

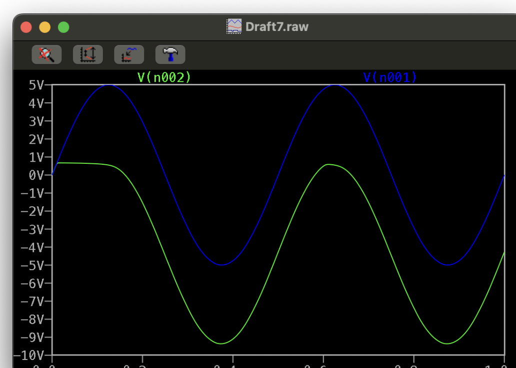
Experiment 3:

1)

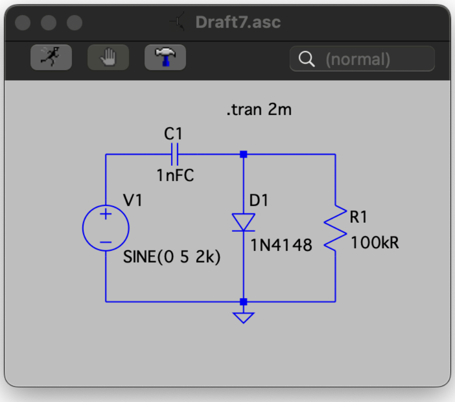


←Figure 5 (100nF)

Figure 6 →

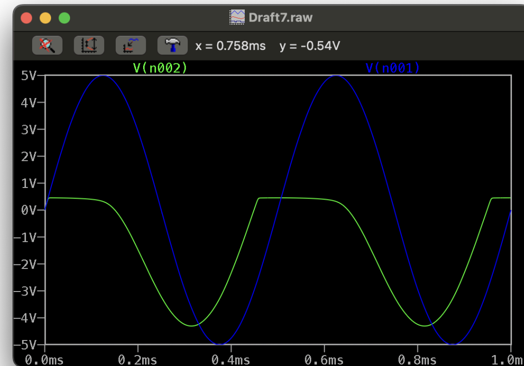


2)



←Figure 7
(1nF)

Figure 8 →



3) The circuit with the 1nF (**Figure 7**) capacitor is less effective at holding the peak value, while the circuit with the 100nF (**Figure 5**) capacitor is better at holding the peak value for a longer time, which could be more effective if we want to maintain the peak longer.